

Multiple Choice (10 marks)

1. Supposing that the clonal selection theory is correct, how can an organism genetically make such diverse proteins?
 - (a) Through vertical gene transfer from parent to offspring
 - (b) Through random assortment of genes during meiosis
 - (c) Through peptide bond formation errors during translation
 - (d) Through genetic mutation only
 - (e) Through transcriptional errors in RNA processing
2. Convergent evolution is best exemplified by which of the following?
 - (a) The gills of a fish and the lungs of a human
 - (b) The leaves of an oak tree and the spines of a cactus
 - (c) The eyes of a human and the eyes of a squid
 - (d) The roots of a tree and the legs of an animal
 - (e) The presence of a notochord in all chordate embryos
3. Which of the following pieces of evidence most strongly supports the common origin of all life on Earth?
 - (a) All organisms reproduce.
 - (b) All organisms are capable of evolving.
 - (c) All organisms require energy.
 - (d) All organisms have DNA.
 - (e) All organisms are composed of cells.
4. What function does oxygen serve in the body?
 - (a) Oxygen is used to maintain the temperature of cells.
 - (b) Oxygen is used by cells to produce alcohol.
 - (c) Oxygen is used to produce carbon dioxide in cells.
 - (d) Oxygen is solely responsible for stimulating brain activity.
 - (e) Oxygen is used to synthesize proteins in cells.
5. What are some of the cues used by animals in migrating?
 - (a) Animals use the temperature changes as cues.
 - (b) Animals use the color of the surroundings as cues.

- (c) Animals use the noises from the environment as cues.
- (d) Animals navigate by following other migrating animal species exclusively.
- (e) Animals rely solely on the memory of previous migration paths.

True / False (10 marks)

Answer True or False

- 6. True or False: The offspring of a person with O type blood and a person with AB type blood will be 50% A type and 50% B type.
- 7. True or False: Guttation is caused by external weather conditions, such as high humidity and low temperatures.
- 8. True or False: The enzyme's constant activity across pH values indicates that only ionizing groups on the substrate participate in the reaction.
- 9. True or False: The probability of drawing the ace of diamonds from a standard deck of cards is $1/53$.
- 10. True or False: Lake B has a higher pH than Lake A, indicating it is more alkaline than Lake A.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the methods you would employ to determine whether a newly discovered bioluminescent worm is the source of light production, rather than a contaminating bacteria.
- 12. Discuss the adaptations that enable whales to thrive in cold water environments, focusing on their thermoregulation and insulation mechanisms.

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